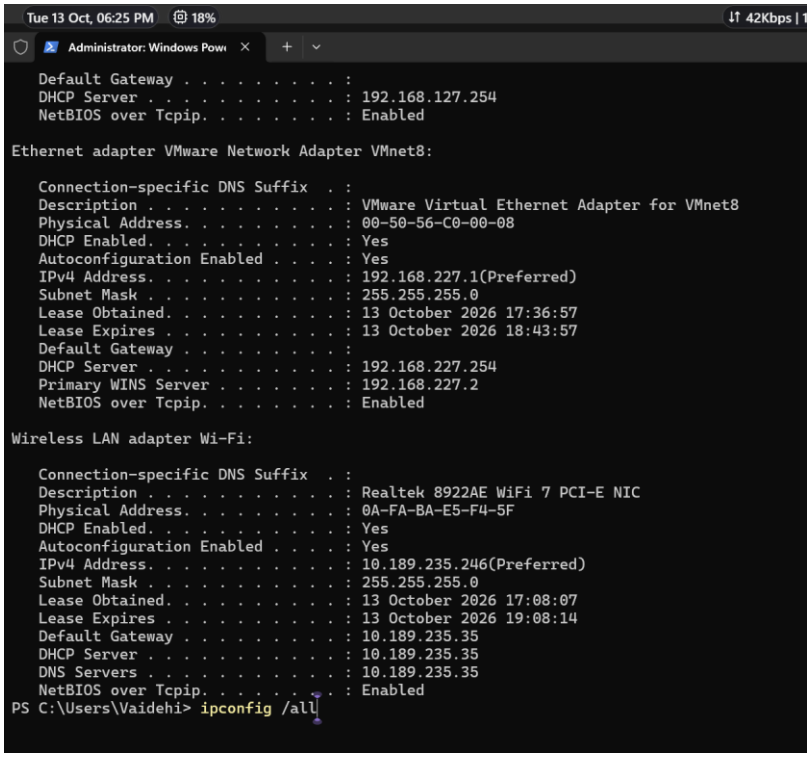
Network Troubleshooting, Management, Monitoring, and Optimization

Q1. Use the Windows command prompt to run the command ipconfig /all on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

Solution : Find network configuration using ipconfig /all



IPv4 Address : 10.189.235.246

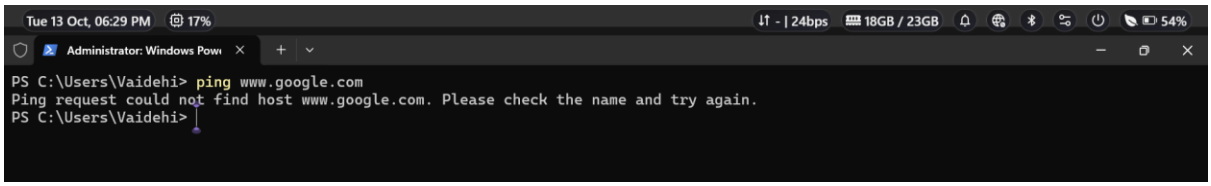
Subnet Mask : 255.255.255.0

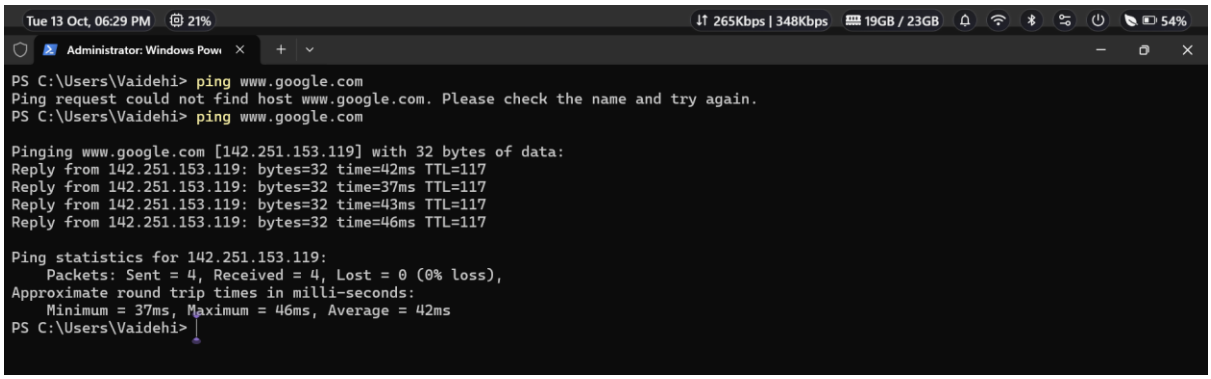
Default Gateway: 10.189.235.35

DNS Servers : 10.189.235.35

Q2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

Solution : Test connectivity with Wi-Fi disconnected and connected

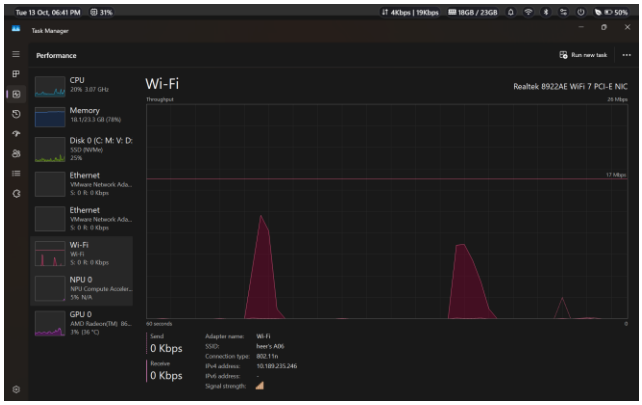




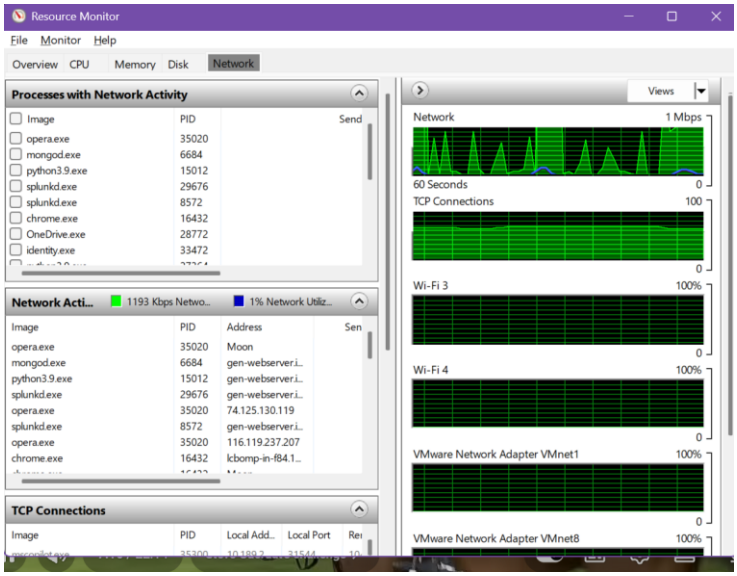
When disconnected, the ping command may fail because the computer cannot reach the destination. After reconnecting, successful replies indicate that connectivity has been restored.

Q3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

Solution : Monitor network usage while streaming YouTube







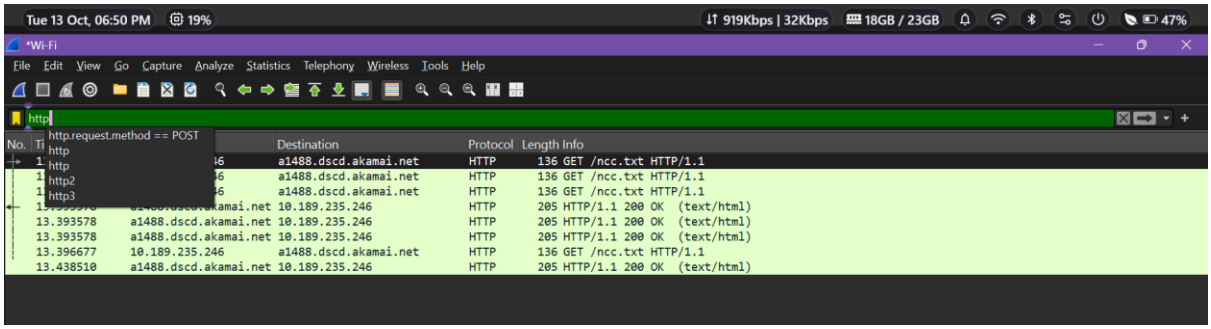
Observation

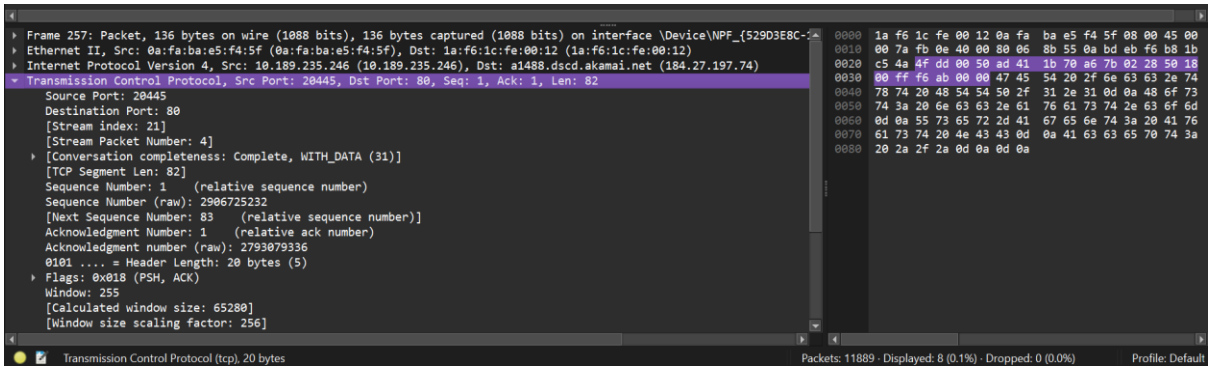
Parameter	Observed Result
Activity	Network usage monitoring
Monitoring tool	Task Manager and Resource Monitor
Network adapter	Wi-Fi
Peak network usage	Approximately 1.2 Mbps (Resource Monitor screenshot)
Highest-usage application	Opera, based on the visible network activity

Task Manager and Resource Monitor help monitor network utilization and identify applications generating network traffic. The screenshots show fluctuating Wi-Fi activity, with Opera appearing in the network activity list.

Q4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets. Hint : Use the filter bar in Wireshark and try http or https as the filter keyword.

Solution : Capture and filter network traffic using Wireshark





Three types of information visible

No.	Information	Description
1	Source and destination IP addresses	Shows which devices or servers are communicating.
2	Protocol and port information	Helps identify TCP, TLS, and associated ports.
3	Packet size and timing	Shows packet lengths and when packets were transmitted.

Q5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

Solution : Troubleshooting checklist for slow Internet while gaming

AI-generated checklist

1. Check Internet speed: Run a speed test and compare the results with your ISP plan.
2. Use Ethernet: Connect the gaming computer directly to the router to reduce Wi-Fi interference.
3. Reduce background traffic: Pause downloads, cloud backups, and unnecessary streaming.
4. Restart the router: Restart the router if the connection is unstable.
5. Check ping and packet loss: Use ping to test latency and connection stability.
6. Improve Wi-Fi placement: Keep the router in an open, central location and use the 5 GHz band when suitable.
7. Update network drivers: Install current network adapter drivers and router firmware.
8. Check server selection: Choose a nearby gaming server to reduce latency.

Suggestion I would try first

I would first connect the gaming computer to the router using an Ethernet cable. This can reduce wireless interference and improve connection stability, especially when multiple devices are using Wi-Fi.